

Supplementary Material

Active phases for high temperature Fischer-Tropsch synthesis in the silica supported iron catalysts promoted with antimony and tin

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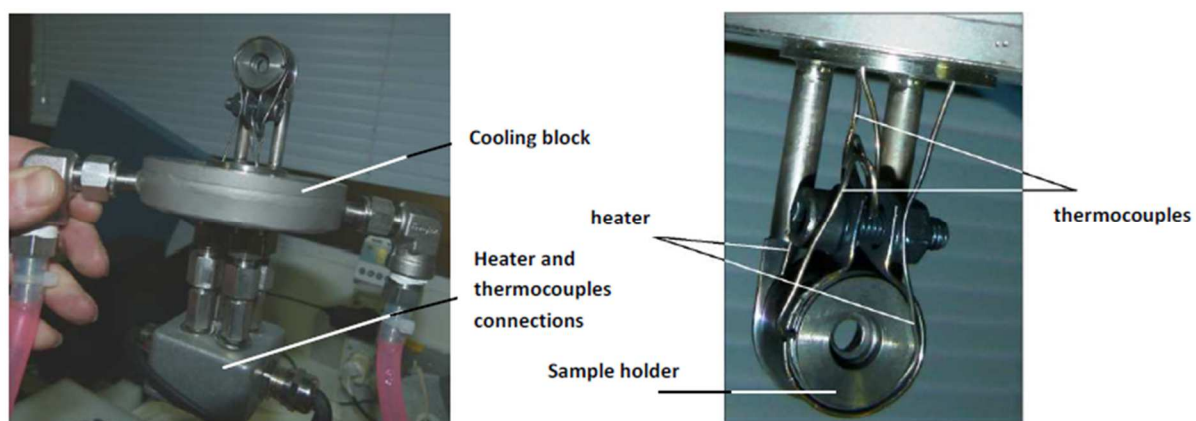
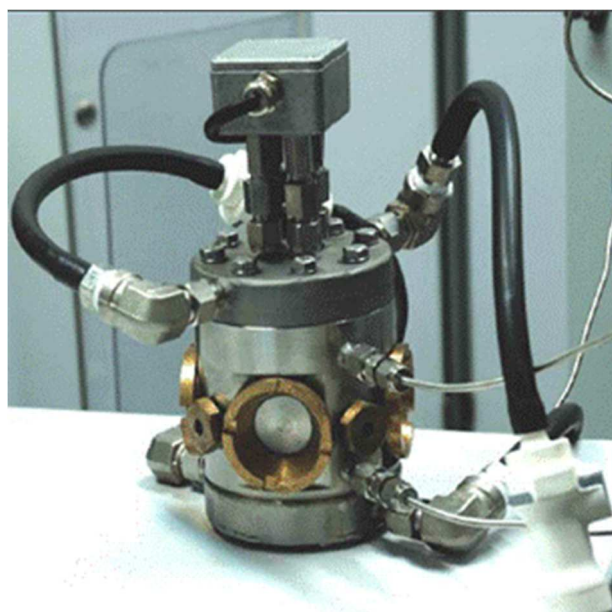


Figure S1. ITQ-ALBA Multipurpose Cell

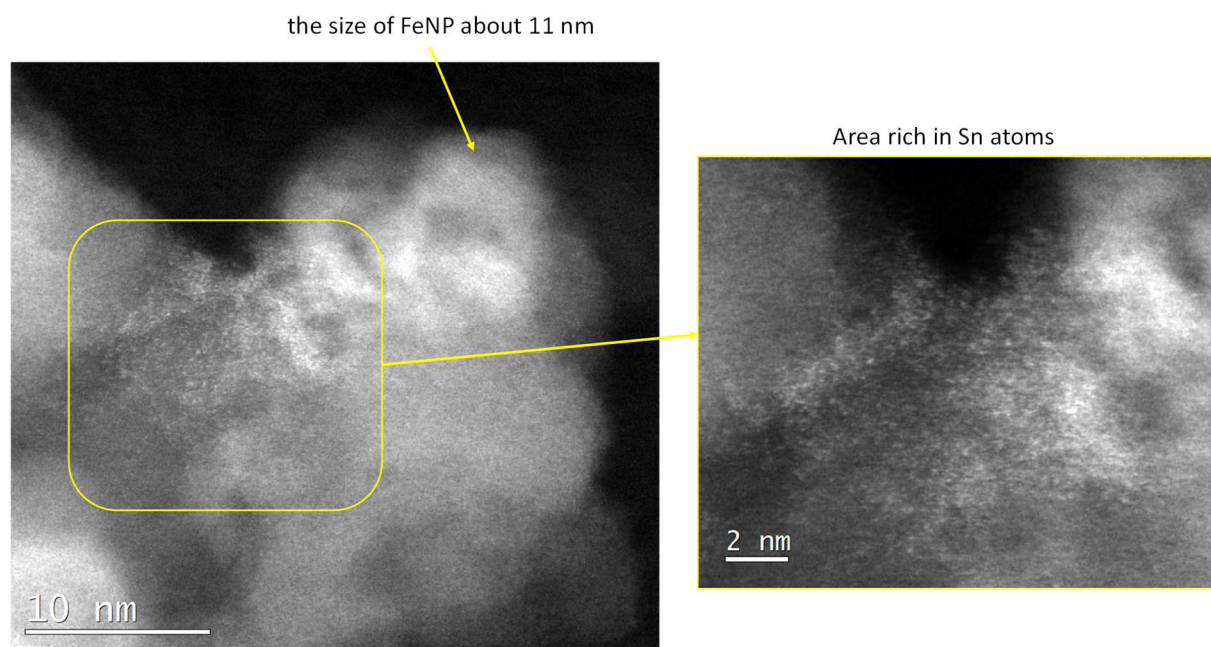
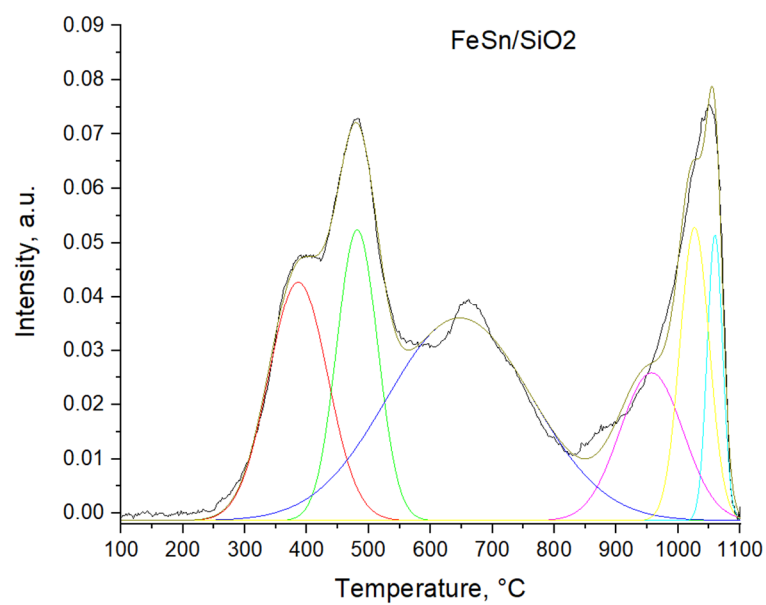
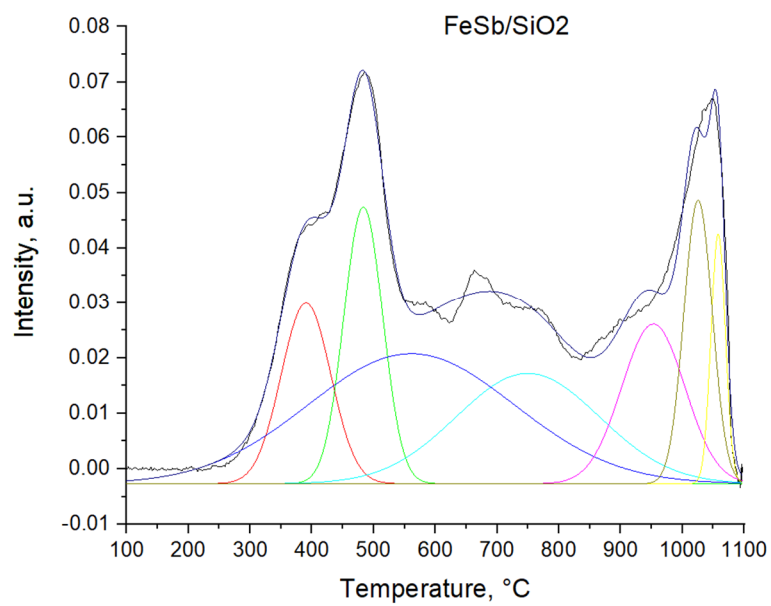


Figure S2. Tin-enriched areas in the close proximity of iron nanoparticles in the activated FeSn/SiO₂ catalyst



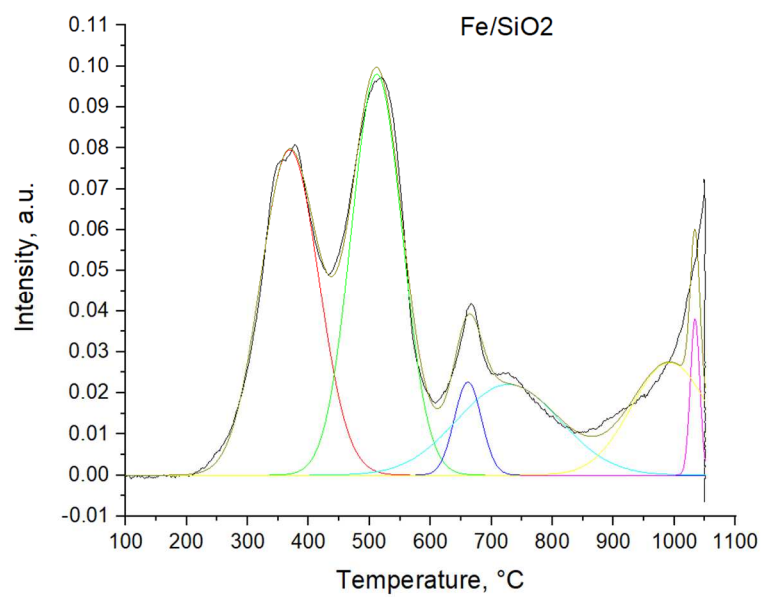


Figure S3. Deconvolution of TPR profiles

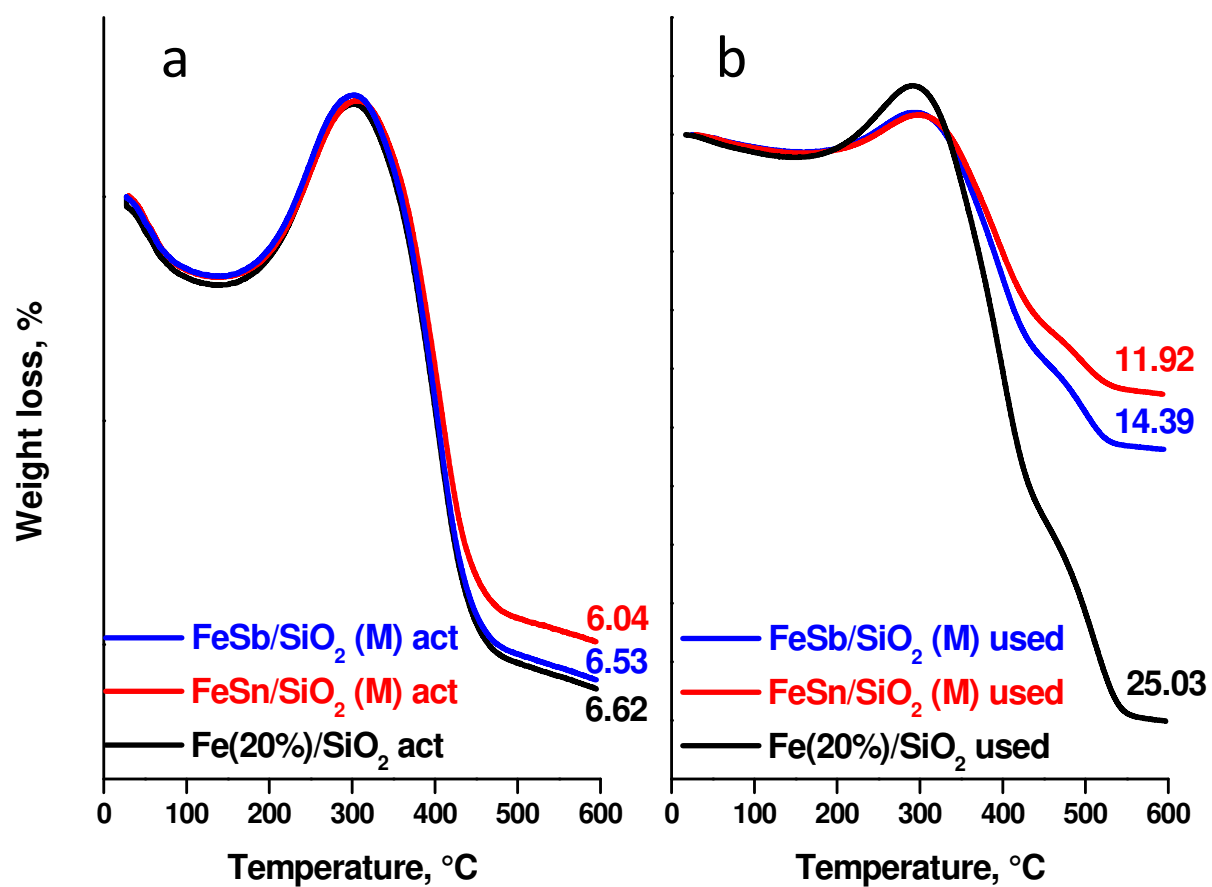


Figure S4. Thermogravimetric analysis (TGA) for catalysts prepared by mechanical mixing

Mössbauer spectroscopy is very sensitive (measuring nuclear energy levels) so different spectral contributions can be observed for individual crystallographic sites of the measured structures. The fresh samples were measured at room temperature (**Figure S4**) revealing the presence of hematite species in all catalysts which agrees with XRD results and EXAFs data. After the CO activation treatment at 350 °C (**Figure S5**) the presence of the active Hägg carbide and the intermediate reduction wüstite species was observed (**Table S1**) for all catalysts excepting FeSb/SiO₂ in which reduction goes further until metallic iron (about 10%).

Table S1. The Mössbauer fitted parameters of activated catalysts, measured at -153 °C

<i>Sample/ Treatment</i>	<i>IS (mm·s⁻¹)</i>	<i>QS (mm·s⁻¹)</i>	<i>Hyperfine field (T)</i>	<i>Γ (mm·s⁻¹)</i>	<i>Phase</i>	<i>Spectral contribution (%)</i>
Fe/SiO ₂ CO, 350 °C	0.26	-	24.3	0.58	χ-Fe ₅ C ₂ (I)	27
	0.20	-	20.3	0.58	χ-Fe ₅ C ₂ (II)	26
	0.19	-	13.2	0.58	χ-Fe ₅ C ₂ (III)	12
	1.23	-0.66	36.0	0.64	Fe _{1-x} O (I- Fe ²⁺)	7
	1.25	-0.27	30.8	0.64	Fe _{1-x} O (II- Fe ²⁺)	14
	0.89	2.11	-	0.82	Fe _{1-x} O (SPM ^a)	14
FeSb/SiO ₂ CO, 350 °C	0.25	-	24.5	0.52	χ-Fe ₅ C ₂ (I)	37
	0.21	-	19.9	0.52	χ-Fe ₅ C ₂ (II)	31
	0.19	-	12.8	0.52	χ-Fe ₅ C ₂ (III)	21
	0.00	-	34.0	0.45	Fe ⁰	11
FeSn/SiO ₂ CO, 350 °C	0.26	-	24.2	0.77	χ-Fe ₅ C ₂ (I)	18
	0.24	-	19.3	0.77	χ-Fe ₅ C ₂ (II)	14
	0.21	-	12.7	0.77	χ-Fe ₅ C ₂ (III)	7
	0.35	0.28	48.6	0.78	Fe _{1-x} O (I- Fe ³⁺)	5
	1.30	-0.65	34.6	0.78	Fe _{1-x} O (II- Fe ²⁺)	20
	1.13	-0.05	30.0	0.78	Fe _{1-x} O (III- Fe ²⁺)	23
	0.98	2.12	-	0.80	Fe _{1-x} O (SPM)	13
20% Fe/SiO ₂ CO, 350 °C	0.27	-	24.0	0.56	χ-Fe ₅ C ₂ (I)	31
	0.19	-	20.2	0.56	χ-Fe ₅ C ₂ (II)	27
	0.18	-	13.4	0.56	χ-Fe ₅ C ₂ (III)	13
	1.16	-0.81	34.5	0.48	Fe _{1-x} O (I- Fe ²⁺)	4
	1.25	-0.36	30.0	0.48	Fe _{1-x} O (II- Fe ²⁺)	11
	0.87	2.11	-	0.70	Fe _{1-x} O (SPM)	14
FeSb/SiO ₂ (M) CO, 350 °C	0.27	-	24.1	0.51	χ-Fe ₅ C ₂ (I)	30
	0.21	-	19.7	0.51	χ-Fe ₅ C ₂ (II)	27
	0.21	-	12.9	0.51	χ-Fe ₅ C ₂ (III)	15
	1.17	-0.21	27.6	0.64	Fe _{1-x} O (Fe ²⁺)	10
	0.89	2.12	-	0.81	Fe _{1-x} O (SPM)	18
FeSn/SiO ₂ (M) CO, 350 °C	0.27	-	24.3	0.53	χ-Fe ₅ C ₂ (I)	27
	0.20	-	20.4	0.53	χ-Fe ₅ C ₂ (II)	29
	0.18	-	13.3	0.53	χ-Fe ₅ C ₂ (III)	14
	1.24	-0.71	35.2	0.54	Fe _{1-x} O (I- Fe ²⁺)	4
	1.26	-0.28	30.3	0.54	Fe _{1-x} O (II- Fe ²⁺)	13
	0.89	2.13	-	0.74	Fe _{1-x} O (SPM ^a)	13

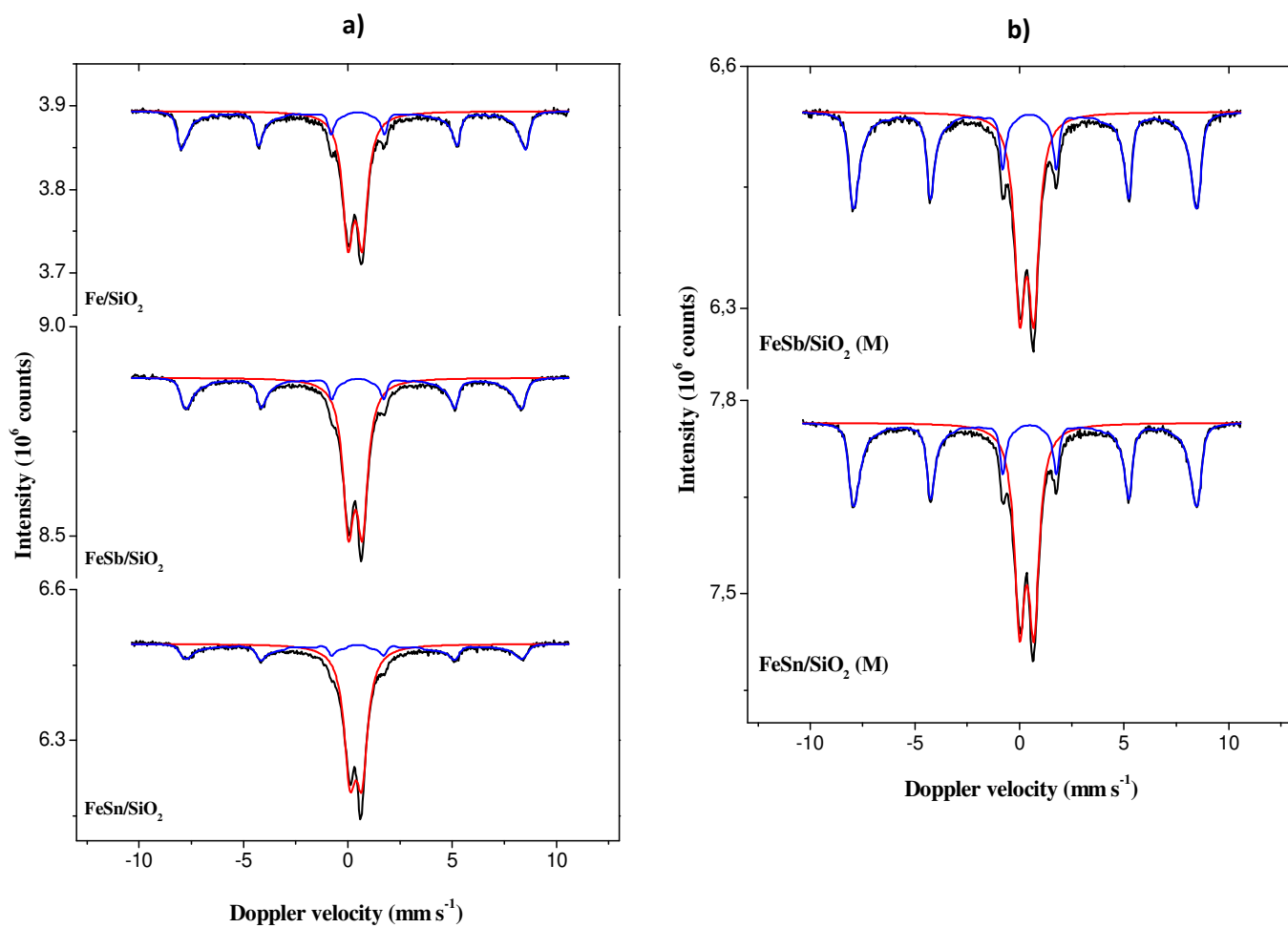


Figure S5. Mössbauer spectra obtained for fresh (a) impregnated and (b) mixed catalysts measured at room temperature

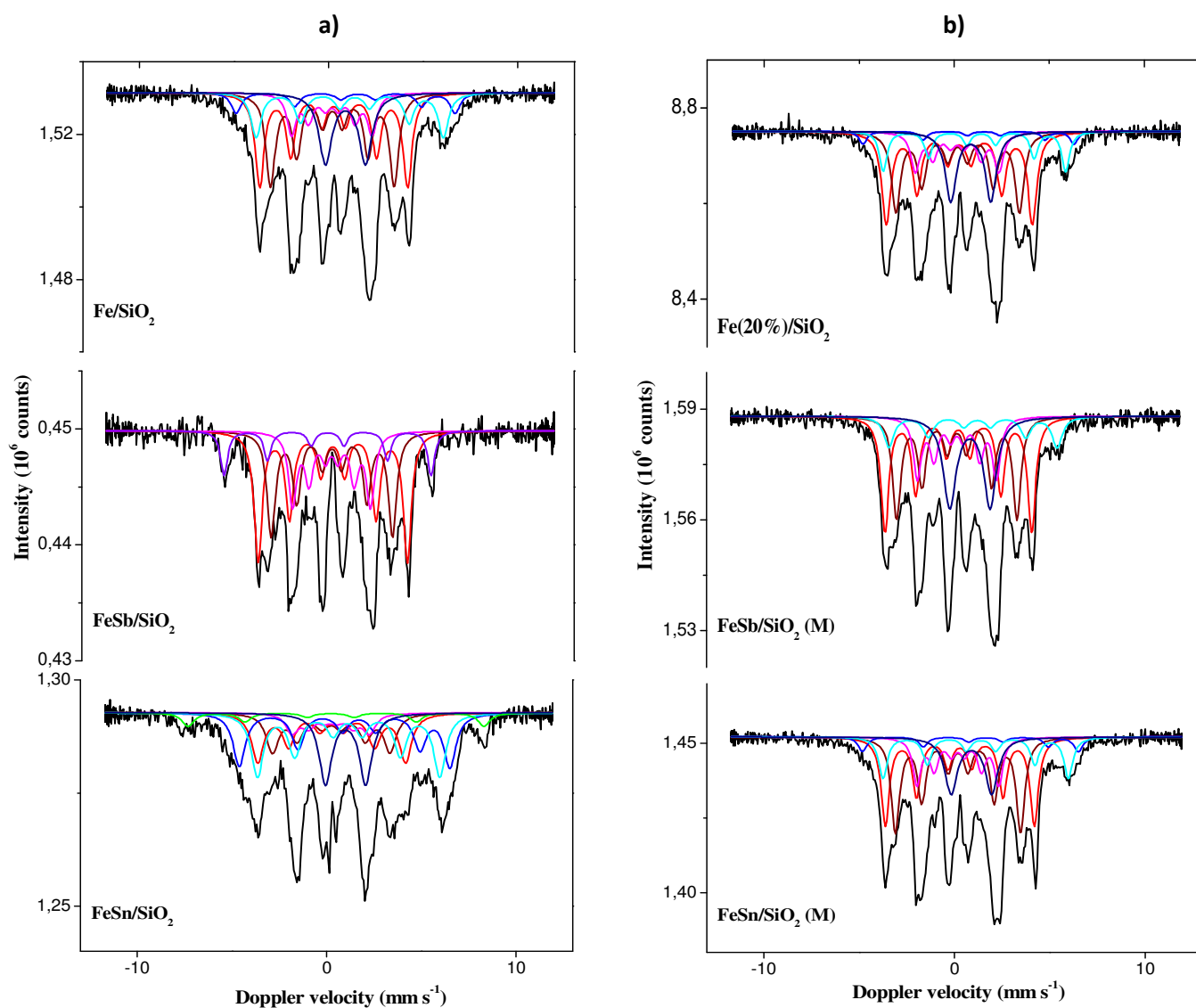


Figure S6. Mössbauer spectra obtained for (a) impregnated and (b) mixed catalysts after CO treatment at 350 °C measured at room temperature

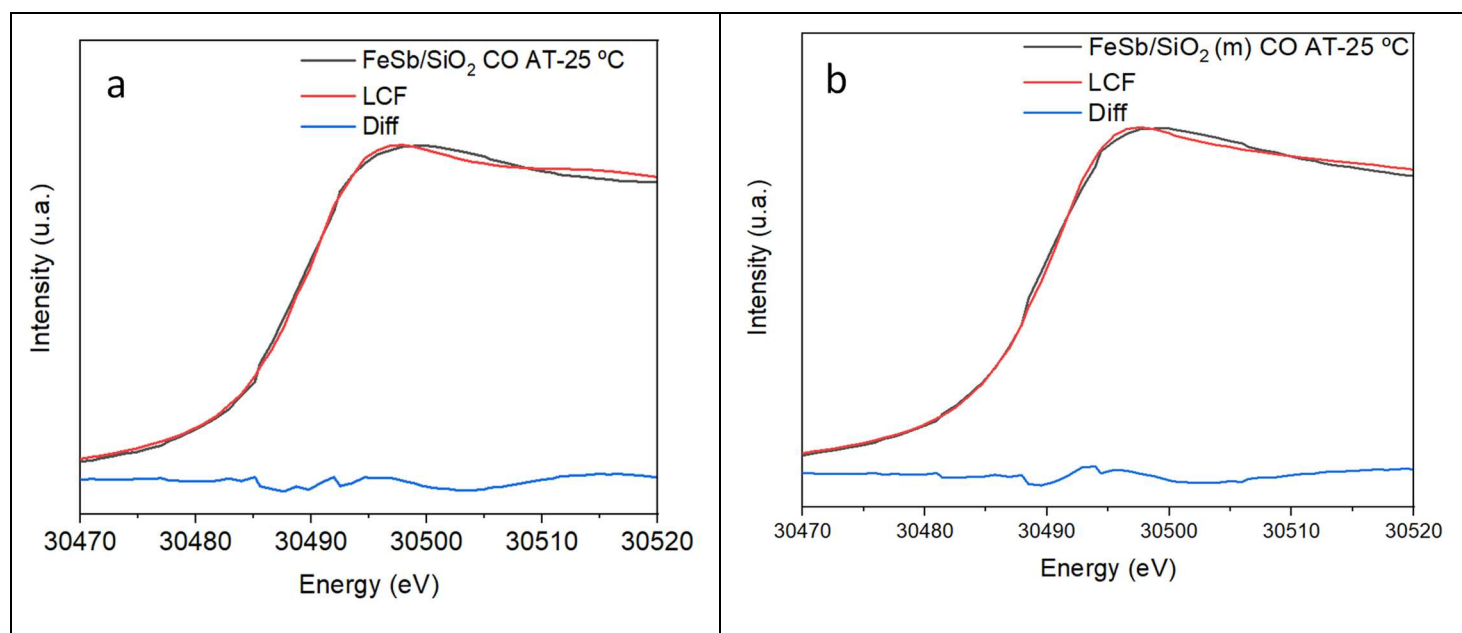


Figure S7. Linear combination fitting and difference spectra for the catalysts after activation and exposure to syngas at 350°C: a- FeSb/SiO₂ ; b – FeSb/SiO₂. The spectra are measured in CO at room temperature.

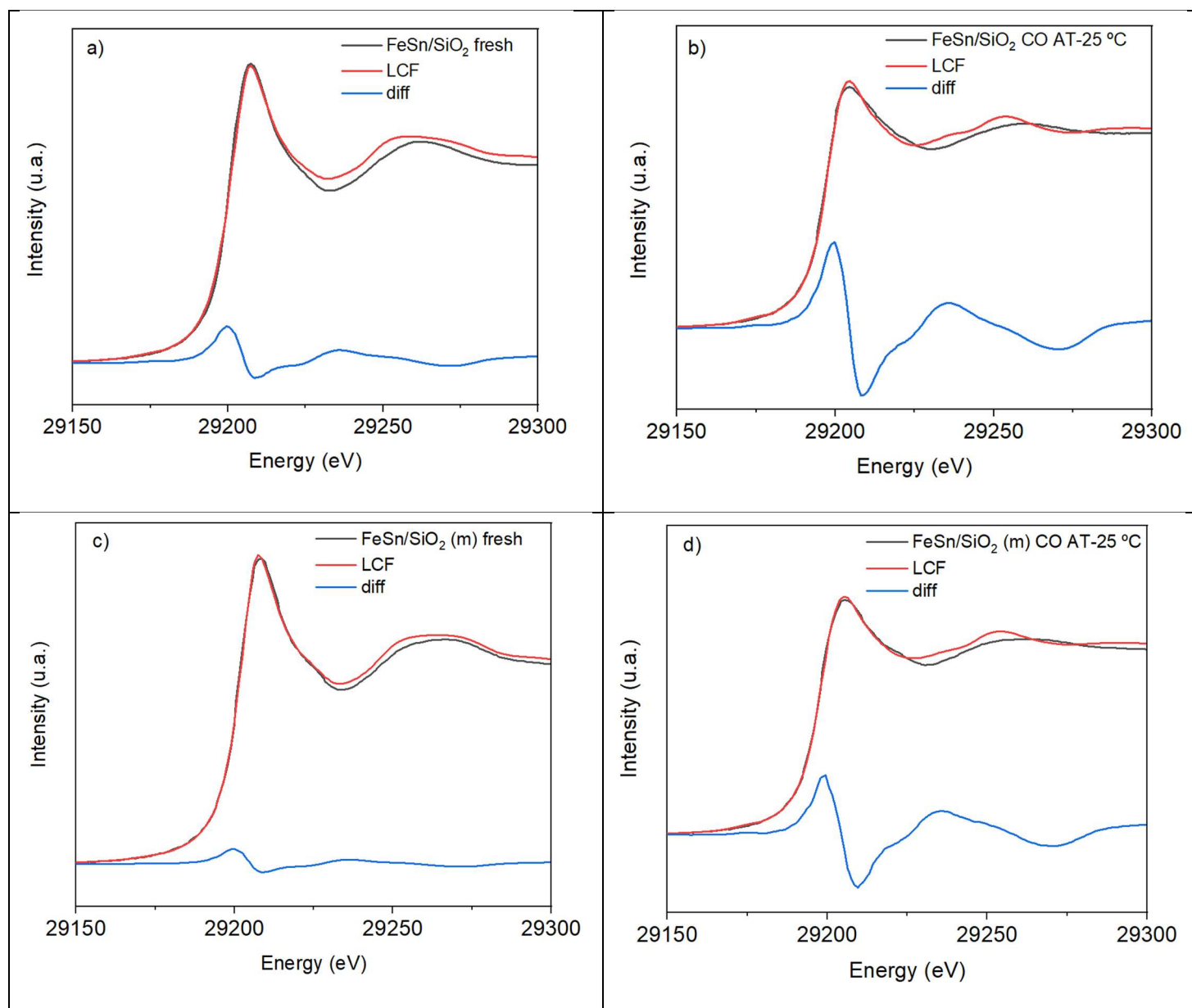


Figure S8. Linear combination fitting and difference spectra: a- calcined FeSn/SiO₂; b – FeSn/SiO₂ after activation and exposure to syngas at 350°C measured CO at 25°C), c- calcined FeSn/SiO₂ (c) ; d – FeSn/SiO₂ (m) after activation and exposure to syngas at 350°C measured CO at 25°C